

Research on AI for autism brain scan diagnosis has a pathologization problem

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Abstract. *Substantial efforts in recent research are directed to the development of Artificial Intelligence (AI) models to support autism diagnosis and biomarker discovery. However, this calls for an important question: does the ongoing research respect the ethical demands of autistic stakeholders themselves? Within that context, this paper examines a strange and alarming phenomenon: the use of the word "disease" to refer to autism. In a scientometric analysis over peer-reviewed machine learning papers in the citation network of the popular neuroimaging dataset ABIDE I, more than half of all papers openly call autism a disease, raising concerns around the values embedded in that research and potential harmful uses of these technologies in the future.*

1. Introduction

Autism is a complex and heterogeneous neurodevelopmental disability characterized by restricted and repetitive behavior patterns and difficulties with social interaction. There are no widely accepted biomarkers for autism and its diagnosis currently relies only on clinical assessment, motivating substantial effort and investment within the scientific community towards the discovery of such biomarkers. A commonly studied approach is to train Machine Learning (ML) models to attempt to diagnose it from biological data, such as Functional Magnetic Resonance Imaging (fMRI), an exam that films the neural activity of each region of the brain over time. However, that body of research has an important ethical problem that needs to be more thoroughly discussed: many of these papers treat autism as a "disease". In this paper, we conduct a scientometric literature review in an attempt to better comprehend the nature and scale of this problem.

2. Background

Using [Pellicano and den Houting 2022] as reference, we introduce the following important concepts, that will be discussed in the paper:

- **Medical model of disability:** under the medical model, disability is a pathological problem within the individual, where the desired outcomes are either cure, prevention, or treatment of "symptoms".
- **Neurodiversity paradigm:** the neurodiversity paradigm rejects the medical model and, instead, sees autism as a natural variation within the diversity of human neurology, with its own strengths and difficulties. As such, rather than a cure, autistic people need accessibility, inclusion, and support.
- **Participatory research:** this kind of research is coproduced with the autistic community, where they may participate in all stages of the research process. This approach usually favors the neurodiversity paradigm and helps ensure that the priorities and outcomes of research align with the demands of autistic stakeholders.

3. Methodology

We conducted a scientometric analysis in the citation network of the ABIDE I (Autism Brain Imaging Data Exchange I) fMRI dataset [Di Martino et al. 2014], one of the most popular open neuroimaging datasets in machine learning for autism biomarker discovery. For reproducibility, the data collected in the analysis and the code are available in a GitHub repository¹.

The inclusion criteria were: **1)** the paper is written in English; **2)** The paper cites the ABIDE I paper; **3)** The title explicitly mentions autism, either with "autism" or "ASD" (Autism Spectrum Disorder); **4)** The paper includes the terms "artificial intelligence", "machine learning", "deep learning" or "neural network" in either the title or the abstract, indicating that Machine Learning is likely to be meaningfully discussed in the paper; and **5)** The paper was published before 2026, to assure, as much as possible, consistency and reproducibility in our search. On May 5th, 2026, we searched for the following query on Scopus:

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( REF ( 2-s2.0-84901284227 ) ) AND TITLE ( "ASD" OR "autism" ) AND TITLE-ABS ( ( "artificial intelligence" OR "machine learning" OR "deep learning" OR "neural network" ) ) AND PUBYEAR < 2026
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Use of the word "disease" in AI/ML papers citing ABIDE I

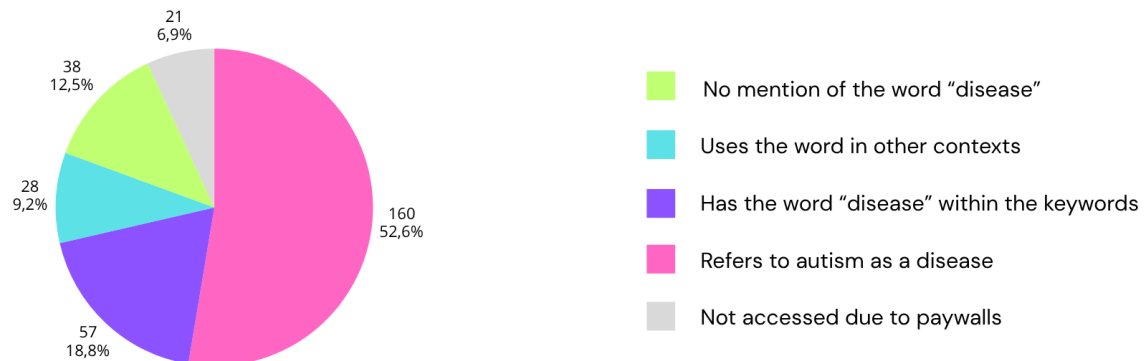


Figure 1. Distribution of pathologization categories in our literature review.

That search produced 304 results. Afterwards, we started searching specifically for papers that mention the word "disease". First, we added AND "disease" to the end of our query, and the new query (on the same day) yielded 266 results. Later, we looked at where and how the term was used in each paper. Among these, 21 papers could not be consulted due to paywalls; hence, we have no data for pathologizing language in them. 28 used the term "disease" in contexts not directly related to purposeful pathologization of autism, such as discussion about Alzheimer's disease, citing another paper that contains the word "disease", or otherwise not having any mention of the term that we could find;

¹<https://anonymous.4open.science/r/Pathologization-in-AI-for-fMRI-autism-diagnosis-5D8B/README.md>

this does not mean that the paper did not have pathologizing language otherwise. 57 did not refer to autism as a disease in the text of the paper, as in the previous case, but used terms such as “diseases” or “brain diseases” as keywords within the Scopus database, which may or may not be intended by the authors. Lastly, 160 refer to autism as a disease. The word was used in many different forms throughout the papers, ranging from (comparably) more subtle forms such as “deep learning is a promising tool to diagnose brain diseases” — implicitly including autism as a disease, given that all papers had “autism” or “ASD” in their title — to more direct uses such as “ASD is a neurodevelopmental disease [...]”. We provide a visualization of this analysis in Figure 1.

After that, we analyzed statistics on the papers labeled as using the word “disease” to refer to autism. In Figure 2, we include visualizations for the Top 10 most frequent categories in **a)** funding agencies and their country of origin; **b)** publication source (e.g. journals or conference proceedings where the papers were published) and their respective publishers; and **c)** country of origin, where, for each country, we counted how many papers had at least one author affiliated with an institution from said country.

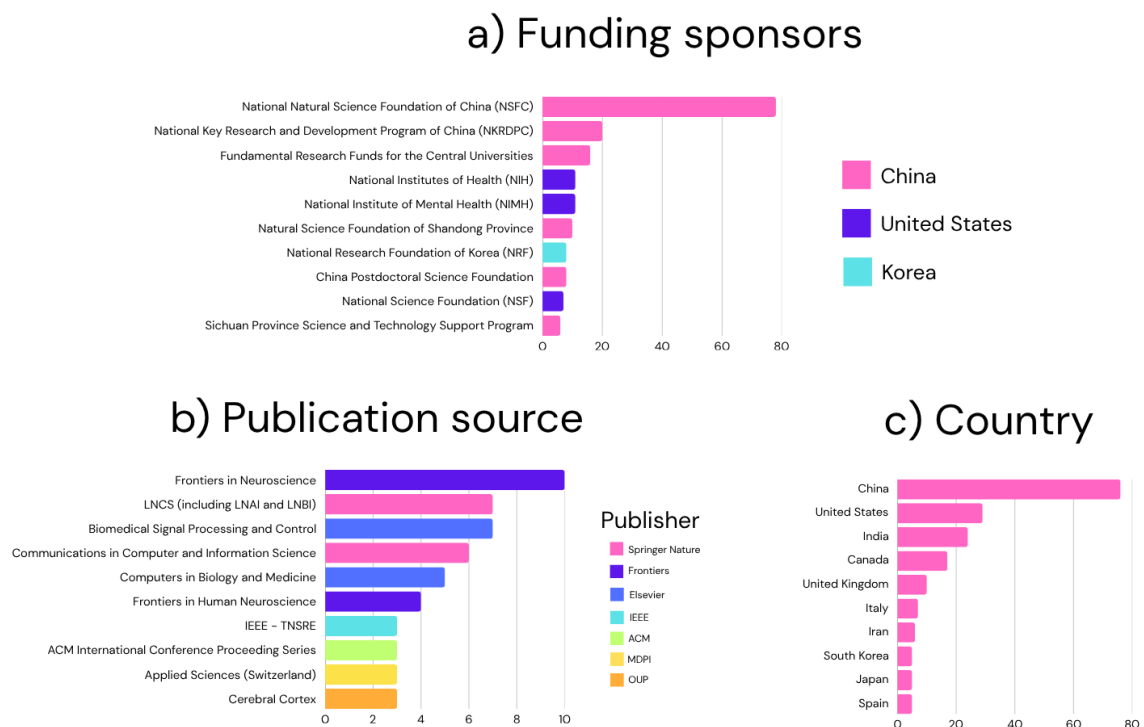


Figure 2. Statistics on papers that referred to autism as a disease.

4. Discussion

Our analysis reveals that much of the research on machine learning for neuroimaging autism diagnosis uses deeply pathologizing vocabulary to refer to autism. These results are likely due to the historical ties between biomedical research and the medical model of disability, as well as the lower engagement with the perspectives of the autistic community and the absence of a participatory research culture [Heraty et al. 2023]. The problem is amplified in machine learning research: while our own review showed that this subfield is

heavily grounded in the medical model, the literature review in [Maye and Hansen 2025] did not find any projects on diagnostic machine learning models following participatory methodologies, despite it being a very active area of research. Another important factor is that much of the research (although not all of it) was developed in Asia, especially China. This reflects not only the high volume of scientific production in those countries in comparison with the rest of the world, but also cultural differences: while current discussions on neurodiversity paradigms are usually centered in the western context, they have emerged only more recently in eastern cultures; also, misunderstandings can happen due to cross-cultural translation [Hirota et al. 2024].

The use of pathologizing vocabulary, besides promoting ableism in public discourse, can actually shape the purposes for which diagnostic models are meant to be used in the future. Although this was not exhaustively explored in our quantitative analysis, these papers often justify the social relevance of their research with the potential to enable early diagnosis and intervention. We do not claim that such interventions are always harmful or unethical; however, under the assumption that autism is a “disease”, early intervention aims to exploit the plasticity of early neurodevelopment to “cure” or “prevent” the development of autism and its “symptoms”, as endorsed in [Dawson 2008]. There is evidence in the literature suggesting that autistic people are generally not supportive of these outcomes [Kapp et al. 2013, Fletcher-Watson et al. 2017].

5. Conclusions and future directions

In this paper, we conducted a scientometric literature review on the ABIDE I citation network and revealed that a considerable and influential body of research on machine learning for neuroimaging diagnosis calls autism a “disease”. We discussed possible causes of the problem, including the prevalence of the medical model and the lack of participatory research, and potential harmful consequences of that discourse in the future.

Future research should be aimed at providing a deeper comprehension of the phenomenon that we observed by uncovering the causes, investigating if and how it manifests in other contexts (such as with other types of biological data or with different datasets), and understanding the practical social consequences; aspects traditionally related to Responsible AI, such as privacy, fairness, accountability and explainability, should also be studied with that problem in mind in the future. We furthermore believe that it is necessary to discuss the adoption of stricter ethical regulations over aspects that enable the development of that research, especially funding, publication criteria, and access to training data. Following principles of participatory research, discussions and decision-making processes on all of these matters should actively include autistic people from diverse backgrounds, as well as other relevant stakeholders such as family members. All of these measures will be important to assure that future research in the field provides beneficial outcomes to autistic people, rather than harm.

6. Ethical considerations

6.1. Ethical Considerations Statement

Although our study uses only publicly available data (curated with a commercial search engine), it manages a relatively delicate discussion where we raised ethical concerns and

criticism of prior scientific work. As such, we do not mean to promote or encourage targeted personal attacks against the authors of the papers we criticized or any other involved parties.

6.2. Generative AI Usage Statement

We did not use Generative AI to prepare this manuscript.

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